Lab 55: The search timed out: tell the coordinator everything

Topic 5: Context Management & Reliability · Task 5.3: Implement error propagation strategies across multi-agent systems

Runs on Simulator 3: Multi-Agent Research System · Run in class

Source: Official exam guide, sample question 8

Commit to an answer first

The web search subagent times out while researching a complex topic. You need to design how this failure information flows back to the coordinator agent. Which error propagation approach best enables intelligent recovery?

A. Return structured error context to the coordinator including the failure type, the attempted query, any partial results, and potential alternative approaches.

B. Implement automatic retry logic with exponential backoff within the subagent, returning a generic "search unavailable" status only after all retries are exhausted.

C. Catch the timeout within the subagent and return an empty result set marked as successful.

D. Propagate the timeout exception directly to a top-level handler that terminates the entire research workflow.

Simulate the solution

```
# In the browser: open Simulator 3, then Lab 55
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 55

# In a terminal, from simulators/simulator-3-multi-agent-research/
uv run research_system.py
```

What you should see

The run finishes, and the report names visual arts as a coverage gap.

Open `simulators/simulator-3-multi-agent-research/evidence/` and find it yourself.

The answer

A. Failure type, attempted query, partial results, alternatives.

Structured error context gives the coordinator the information it needs to make intelligent recovery decisions—whether to retry with a modified query, try an alternative approach, or proceed with partial results. Option B's generic status hides valuable context from the coordinator, preventing informed decisions. Option C suppresses the error by marking failure as success, which prevents any recovery and risks incomplete research outputs. Option D terminates the entire workflow unnecessarily when recovery strategies could succeed.

Now change one thing

In fixtures/sources.json replace the visual-arts TIMEOUT with [] and rerun: a valid empty result must not read as a failure.